

Multifractality as a sensing resource: a D_2 figure of merit for Rosenzweig–Porter systems, and a lattice platform

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Abstract

Two established results have never been composed. The fidelity susceptibility of a quantum state to a Hamiltonian parameter equals one quarter of its quantum Fisher information, $F_Q = 4\chi_f$ [1, 2]; and an exact calculation in the Rosenzweig–Porter (RP) ensemble shows that eigenfunction fidelity susceptibility is *enhanced* in the fractal non-ergodic-extended phase [4]. Composed, they make the eigenvector fractal dimension D_2 — until now a diagnostic of localization physics — an operational **metrological figure of merit**: along the RP fractal phase, at fixed distance from criticality and for field-like (local) couplings to the sensed parameter, the finite-size enhancement exponent of the quantum Fisher information over the ergodic reference is $1 - D_2$. We confirm this in-house under preregistered criteria on the generalized-RP ensemble with D_2 *measured* from inverse-participation-ratio scaling (not assumed): enhancement = $1 - D_2$ within 0.15 at 4/4 interior points of the phase (deviations 0.009–0.078; correlation 0.996). Two further preregistered experiments fix the criterion’s boundaries: geometric fractality alone — a disorder-free Sierpinski-gasket lattice with certified eigenpairs and degenerate-multiplet treatment — shows **no effect** (enhancement -0.12 , $|\cdot| < 0.15$; the RP formula does not extend), while critical power-law banded random matrices **extend the law** at all three tested bandwidths (deviations 0.016–0.064). The criterion is therefore: *multifractal eigenstates of random ensembles, sensed through local couplings* — randomness, not fractal geometry, is the missing ingredient, and an operator-structure scope law (diagonal inherits; short-range inherits once its range fits the fractal support; collective and dense couplings are D_2 -blind) gives the design rule. On the platform side we construct — because it cannot be cited — the minimal model that carries the physics onto cold atoms: an *interacting tilted+disordered lattice* in which the tilt is only the sensed parameter (the bare Wannier–Stark ladder is integrable, so sourcing RP statistics from gravity would be immediately refutable) and the dense RP-like spectrum comes from interactions and disorder. A preregistered coexistence test on the interacting t – V chain finds a textbook ergodic→localized crossover ($\langle r \rangle$: $0.532 \rightarrow 0.386$) whose *typical* susceptibility peaks at the crossover’s ergodic edge — not mid-crossover — with the predicted heavy x^{-2} tail (Hill index 1.96–2.04) in the chaotic regime and estimator-dependent rare-event gains inside the crossover. The platform statement is scoped accordingly, and one instrument caveat is stated up front: this attaches to *lattice/Bloch* gravimetry, not to free-fall light-pulse interferometry. Tier-2 throughout: a shared random-matrix description with preregistered numerical evidence, not a hardware claim.

1 Introduction

Two sibling programs measure the same observable on different substrates. A physics program tests a Rosenzweig–Porter (RP) hypothesis for freely vibrating plates — proposing the hypothesis with a preregistered eigenvector fractal dimension [17] and then executing its registered numerical campaign on true operators [18]. An AI program finds that a neural network’s fabricated Poisson→GOE transition carries genuinely RP-multifractal Hessian eigenvectors, $D_2 = 0.76$ within an architecture band $[0.56, 0.80]$ [19]. In both, D_2 is the *decisive diagnostic*: it separates a genuine multifractal phase from spectra that merely imitate intermediate spacing statistics.

This note asks what D_2 is *for*, beyond diagnosis. The metrology literature has recently begun to name multifractal states “a resource for quantum sensing” [9], and criticality-enhanced sensing is an established framework [10, 12]; but no standalone, quantitative figure of merit tied to the fractal dimension itself has been formulated — the notion is adjacent and nameable, yet the cell is empty. Meanwhile the two ingredients of such a figure of merit have existed, separately, for years: the identity $F_Q = 4\chi_f$ between quantum Fisher information and fidelity susceptibility [1, 2, 3], and Kravtsov and collaborators’ exact result that eigenfunction fidelity susceptibility is enhanced in the RP fractal phase [4] — derived in the same ensemble the plate program uses [5].

We compose them, state the resulting figure of merit operationally, and then do what the parent programs’ discipline requires: preregister the readings, run the tests, and report the boundaries of validity with the same weight as the confirmations. The result is a criterion — multifractal eigenstates of *random* ensembles, sensed through *local* couplings — that survived one designed attempt to break it (a disorder-free fractal lattice, where it correctly reports no effect) and extended without adjustment to a third ensemble family. Part 1 of the contribution constructs the minimal cold-atom platform on which the figure of merit is operational; Part 2 is the figure of merit itself. We present Part 2 first, since it is load-bearing.

2 The figure of merit

Definitions. For a Hamiltonian $H(h) = H + hV$ and eigenstate $|n(h)\rangle$, the fidelity susceptibility is

$$\chi_f^{(n)} = \lim_{h \rightarrow 0} \frac{-2 \ln |\langle n(0) | n(h) \rangle|}{h^2} = \sum_{m \neq n} \frac{|\langle m | V | n \rangle|^2}{(E_m - E_n)^2},$$

and $F_Q = 4\chi_f$ is the quantum Fisher information of the eigenstate as a probe of h [1, 2]: the Cramér–Rao-optimal sensitivity to the parameter, up to a factor fixed by convention. Spectroscopy-accessible formulations of QFI are standard [3].

The exact input. In the RP ensemble — $H = \text{diag}(\epsilon_i) + N^{-\gamma/2}W$ with GOE W — the fractal non-ergodic-extended phase at $1 < \gamma < 2$ has eigenstates occupying N^{D_2} of N available sites, $D_2 = 2 - \gamma$ [5]. Skvortsov, Amini and Kravtsov [4] computed the eigenfunction fidelity susceptibility exactly in this ensemble and found it *enhanced* in the fractal phase: the mini-band structure concentrates spectral weight at small denominators.

The operational statement. Let $\chi_f^{\text{typ}}(N)$ be the typical (log-averaged) mid-spectrum fidelity susceptibility at system size N for a *field-like* perturbation — diagonal in the localization basis, the analogue of a local field or a tilt — and let $\chi_f^{\text{erg}}(N)$ be the same quantity for a dense (GOE-like) perturbation, which we show below is D_2 -blind and therefore serves as the in-situ ergodic reference.

Define the enhancement exponent α_{enh} by $\chi_f^{\text{typ}}/\chi_f^{\text{erg}} \sim N^{\alpha_{\text{enh}}}$. The figure of merit is

$$\alpha_{\text{enh}} = 1 - D_2$$

along the fractal phase, at fixed distance from its edges, with D_2 the *measured* inverse-participation-ratio scaling dimension of the same eigenstates. Smaller fractal dimension — more concentrated states — buys polynomially more Fisher information per added system size, relative to what an ergodic sensor of the same size would gain.

3 Preregistered confirmation on measured D_2

All experiments in this note follow the parent programs’ discipline [18]: readings frozen in the experiment header before the run, verdicts quoted as frozen, amendments dated. The registry and code are in the companion repository.¹

Design (Q1/Q1b). Generalized-RP ensemble at interior points $\gamma \in \{1.2, 1.4, 1.6, 1.8\}$ — each at least 0.2 from both phase edges, so criticality is held at fixed distance — with ergodic ($\gamma = 0.5$) and localized ($\gamma = 2.5$) controls; $N = 256\text{--}4096$; χ_f per mid-spectrum eigenstate for a diagonal (field-like) perturbation and for a normalized GOE perturbation; D_2 measured from IPR scaling, never assumed equal to $2 - \gamma$.

Results. Measured $D_2 = 0.854/0.608/0.397/0.168$ against the theoretical $0.8/0.6/0.4/0.2$ (agreement 0.03–0.05). The diagonal-coupling exponent obeys $\alpha_{\text{diag}} = 2 - D_2$ within 0.03–0.06, while the GOE-coupling exponent is flat at $\alpha_{\text{GOE}} \approx 1.00$ across the entire phase: **a dense perturbation is D_2 -blind**, which is what licenses it as the in-situ ergodic reference. The enhancement exponent $\alpha_{\text{diag}} - \alpha_{\text{GOE}}$ equals $1 - D_2$ within 0.06 at all four interior points. A second, fully frozen-gate run on an independent seed (Q1b; gate $|\alpha_{\text{enh}} - (1 - D_2)| \leq 0.15$ at $\geq 3/4$ points, declared before the run) passes 4/4 with deviations 0.015/0.018/0.009/0.078 and $\text{corr}(\alpha_{\text{enh}}, 1 - D_2) = 0.996$; monotonicity across the phase passes with total variation 0.709. One scoping finding emerged and is promoted to part of the claim: *the figure of merit requires field-like coupling to the sensed parameter* — it is a statement about how multifractal support meets a local operator, not about the spectrum alone.

4 The boundaries: what multifractality is not enough

Geometric fractality alone fails (Q3b). The sharpest referee question the composition faces [4, 8] is whether “multifractality helps” is generic. We built the adversarial test: a disorder-free Sierpinski-gasket tight-binding lattice, whose eigenstates are multifractal from geometry alone [7] — no randomness, no RP mini-bands. Instrumentation matters here and is reported because it changed a verdict: a first run was invalidated by an eigensolver defect (LAPACK `dsyevd` via `numpy` returning eigenvalues off by up to 31 at $N = 9843$) and by a degeneracy-exclusion design that starved the statistics; the certified rerun (independent eigenvalue reference, `evr`-driver vectors, agreement and residual gates per size, and degenerate-multiplet subspace rotation — correct degenerate perturbation theory) passes certification at all four sizes with full-window statistics. Frozen verdict: enhancement -0.123 , i.e. **no effect** ($|\cdot| < 0.15$), and the RP formula does not extend (deviation 0.816 from $1 - D_2$). The robust statement, given the gasket’s composition scatter, is the absence of any consistent enhancement.

¹`rp-qfi-sensing`; experiments q01–q05. Release plans for the repository accompany the deposit of this note.

A third random family extends the law (Q5). Critical power-law banded random matrices [6] at bandwidths $b = 0.5/1.5/4.0$: $\alpha_{\text{enh}} = 1 - D_2^{\text{meas}}$ within 0.016–0.064 at all three. Combined with Q1b (RP: yes) and Q3b (deterministic gasket: no), the criterion is not “RP specifically” and not “fractality generically” but:

multifractal eigenstates of random ensembles, sensed through local couplings.

What the gasket lacked was randomness — its exact deterministic multiplets carry no mini-band level structure for a local operator to exploit.

The suppression counterexample, reconciled. In the 3D Anderson model, fidelity susceptibility is *submaximal* at the multifractal critical point, and that suppression is attributed to the fractal structure of critical states [8]. This does not contradict the figure of merit; it delimits it. The RP result lives on an extended fractal *phase* at fixed criticality distance; the Anderson result compares a single critical *point* against its own ergodic side. “Multifractality helps” is model-dependent — which is precisely why the criterion above names the random-ensemble phase, not multifractality at large.

5 The operator scope law

Which couplings inherit the enhancement (Q4, frozen classification on gRP at $\gamma \in \{1.4, 1.8\}$):

coupling V	verdict	reading
diagonal (field-like)	inherits	at both γ ; the figure of merit’s home
banded, range 8	conditional	58% at $\gamma = 1.4$; full at $\gamma = 1.8$
low-rank (collective)	blind	no D_2 dependence
dense (GOE-like)	blind	the ergodic reference of §3

A short-range operator inherits the enhancement once its range fits inside the fractal support scale; collective and dense couplings never do. The design rule for any platform is therefore: **sense with local couplings**. (The tilt used in §6 is field-like by construction.)

6 The platform: an interacting tilted+disordered lattice

Why constructed rather than cited. A systematic, adversarially verified literature search (documented in the program’s research record) found each of the bridge’s four required pieces published separately — gravity entering a lattice as a Wannier–Stark tilt with $\partial E_n/\partial g$ a literal eigenvalue derivative [13, 14]; spectroscopic access to QFI [3]; $F_Q = 4\chi_f$ instantiated on tilted and quasiperiodic lattices [10, 11] — but never composed with RP-class level statistics, and with two load-bearing caveats confirmed independently: the bare Wannier–Stark ladder is *integrable* (an equidistant picket fence, the opposite of a dense RP spectrum), and the existing cold-atom QFI enhancements are criticality effects with no level-statistics content. The minimal defensible model is therefore an **interacting, disordered, tilted lattice** (many-body Bose–Hubbard or interacting Aubry–André–Harper class), with a non-negotiable design rule: *the tilt is only the sensed parameter*; the RP-like statistics must come from interactions and disorder, never from gravity.

The coexistence test (Q2). The decisive preregistered question: does such a system show its Poisson $\leftrightarrow$ GOE crossover in the same window where the fidelity susceptibility to the tilt is large? Instantiation: the interacting t - V chain (spinless fermions, $J = V = 1$) with onsite disorder W and the tilt as sensed parameter at $h = 0$; $L = 12$ (Hilbert dimension 924, 200 realizations) and $L = 14$ (3432, 64); first realization of every cell certified against an independent eigensolver. Results: a textbook crossover ($\langle r \rangle : 0.532 \rightarrow 0.386$ as $W : 0.5 \rightarrow 8$); the *typical* susceptibility peaks at $W^* = 2$ where $\langle r \rangle = 0.520$ — the crossover’s **ergodic edge**, not mid-crossover, so the frozen mid-crossover gate reads CHALLENGES and the platform statement is re-scoped accordingly. The predicted GOE x^{-2} heavy tail of the χ_f distribution is confirmed (Hill index 1.96–2.04 across the chaotic regime); *inside* the crossover the tail index drops below 2 (1.73/1.54): the mean susceptibility explodes on rare resonances while the typical value collapses, so sensitivity there is estimator-dependent.

The platform statement (as re-scoped, honestly). Maximum *typical* sensitivity to a field-like parameter sits at the chaotic edge of the crossover; the crossover interior offers heavy-tailed, rare-event gains that a mean-based protocol can harvest and a typical-based protocol cannot. Whether the peak tracks the finite-size crossover point or pins to fixed disorder is registered as the follow-up (Q2c, finer grid, three sizes).

The instrument caveat, stated before it is asked. A light-pulse (Kasevich–Chu) atom interferometer reads gravity from a free-fall phase $\Delta\phi = k_{\text{eff}} g T^2$, which is not a spectral quantity. Everything above lives on *lattice/Bloch gravimetry* — the Wannier–Stark/Bloch-oscillation class of instruments, a cousin of the free-fall interferometer, not its own readout. The bridge from RP structure to a light-pulse phase, if one exists, is future work and is not claimed here.

7 Relation to the parent programs, and outlook

The figure of merit gives the parent programs’ D_2 measurements an operational meaning. The AI program’s fabricated transition ($D_2 = 0.76$, band $[0.56, 0.80]$ [19]) and the designed plate assembly that lands in the same band at router-like coupling ($D_2 = 0.63$ – 0.69 [18]) sit squarely in the regime the criterion names: multifractal states of effectively random couplings. Nature’s free plate, whose genuine true-operator effect is an order of magnitude weaker ($D_2 \approx 0.01$ – 0.25 , ordered by boundary adaptedness [18]), correspondingly offers little of this resource — consistent with the campaign’s central protocol-artifact finding. Enhancement exponents of $1 - D_2 \approx 0.24$ – 0.44 (AI band) versus ≈ 0.3 – 0.4 (designed assemblies) are, on this account, engineering targets rather than diagnostics.

Open, in order of leverage: (i) the Q2c size-tracking question above; (ii) energy-resolved sharpening of the gasket null (registered as Q3c); (iii) an atomic GOE $\rightarrow$ GUE spacing measurement — no atomic system has ever shown that crossover in level statistics, and the mechanical version is now fully characterized in-silico [18]. A dedicated feasibility review (2026-07-11, this program’s research record) finds the atomic kicked-rotor route *not yet* statistically feasible: the bottleneck is resolved-level count (tens of levels demonstrated on the closest platforms, against the ~ 50 – $100+$ a spacing histogram needs), while the two other ingredients exist separately — quantum-resonance delocalization is a proven mechanism, and a genuine time-reversal-control knob has been demonstrated on a cold-atom Floquet platform, though verified only through transport observables, never through quasienergy spectroscopy. No platform yet combines all three; the prediction is thereby a defined engineering gap rather than an open guess, and as of this writing the field is clean — the closest adjacent work computes genuine spacing statistics on a kicked rotor but explicitly remains in the orthogonal class. (iv) The composition claimed in this note — multifractal dimension as a metrological figure of merit — likewise remains unclaimed in the 2024–2026 literature at the time of deposit, though its

ingredients are public and recent [4, 9]; priority is the point of depositing now.

8 Methods and reproducibility

Generalized-RP ensemble $H = \text{diag}(\epsilon_i) + N^{-\gamma/2}W$, ϵ_i i.i.d., W GOE; χ_f summed exactly over the spectrum per mid-window eigenstate; typical values are log-averages; D_2 from IPR ladders over $N = 256\text{--}4096$ with window fractions held fixed. Spacing statistics via the consecutive-spacing ratio [15, 16], per symmetry sector, against finite-size baselines computed with the identical protocol. Eigenpair certification — an independent no-vector eigenvalue reference, driver-level cross-checks, agreement and residual gates per size — is mandatory throughout; the one solver defect found (§4) was caught by these gates and is documented with the fix. Every experiment carries its preregistered reading in the script header; all numbers quoted above are frozen verdicts. Compute: CPU-only commodity workstation.

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